Task - 4: Group Task
Project: Difference between No Uage of Numpy and with Numpy:

1. Adding two arrays

Without NumPy — using a loop

```
[ ]: 1. Adding two arrays (Without Numpy)
```

```
[27]: a = [1, 2, 3, 4]
      b = [5, 6, 7, 8]
      result = []

      for i in range(len(a)):
          result.append(a[i] + b[i])

      print(result)
```

```
[6, 8, 10, 12]
```

With NumPy

```
[ ]: With NumPy
```

```
[28]: import numpy as np

      a = np.array([1, 2, 3, 4])
      b = np.array([5, 6, 7, 8])

      print(a + b)
```

```
[ 6  8 10 12]
```

2. Multiplying every element by 2

Without NumPy

```
[ ]: 2. Multiplying every element by 2
     (Without NumPy)
```

```
[29]: a = [1, 2, 3, 4]
      result = []

      for x in a:
          result.append(x * 2)

      print(result)
```

```
[2, 4, 6, 8]
```

With NumPy

```
[30]: a = np.array([1, 2, 3, 4])
      print(a * 2)
```

```
[2 4 6 8]
```

3. Finding the sum

Without NumPy

```
[ ]: 3. Finding the sum
     Without NumPy
```

```
[31]: a = [10, 20, 30, 40]
      total = 0

      for x in a:
          total += x

      print(total)
```

```
100
```

With NumPy

```
[ ]: With NumPy
```

```
[32]: a = np.array([10, 20, 30, 40])
      print(np.sum(a))
```

```
100
```

4. Finding the average

Without NumPy

```
[ ]: 4. Finding the average
     Without NumPy
```

```
[34]: a = [10, 20, 30, 40]
      total = 0

      for x in a:
          total += x

      print(total / len(a))
```

```
25.0
```

With NumPy

```
[ ]: With NumPy
```

```
[35]: a = np.array([10, 20, 30, 40])
      print(np.mean(a))
```

```
25.0
```

5. Filtering values greater than 5

Without NumPy

```
[ ]: 5. Filtering values greater than 5
     Without NumPy
```

```
[36]: a = [2, 7, 4, 9, 1, 8]
      result = []

      for x in a:
          if x > 5:
              result.append(x)

      print(result)
```

```
[7, 9, 8]
```

With NumPy

```
[ ]: With NumPy
```

```
[37]: a = np.array([2, 7, 4, 9, 1, 8])
      print(a[a > 5])
```

```
[7 9 8]
```

6. Finding the maximum value

Without NumPy

```
[38]: a = [12, 45, 23, 67, 34]
      maximum = a[0]

      for x in a:
          if x > maximum:
              maximum = x

      print(maximum)
```

```
67
```

With NumPy

```
[39]: import numpy as np

      a = np.array([12, 45, 23, 67, 34])
      print(np.max(a))
```

```
67
```

7. Squaring every element

Without NumPy

```
[40]: a = [2, 3, 4, 5]
      result = []

      for x in a:
          result.append(x ** 2)

      print(result)
```

```
[4, 9, 16, 25]
```

```
[ ]:
```

With NumPy

```
[41]: a = np.array([2, 3, 4, 5])
      print(a ** 2)
```

```
[ 4  9 16 25]
```

8. Subtracting two arrays

Without NumPy

```
[42]: a = [10, 20, 30]
      b = [1, 2, 3]
      result = []

      for i in range(len(a)):
          result.append(a[i] - b[i])

      print(result)
```

```
[9, 18, 27]
```

With NumPy

```
[43]: a = np.array([10, 20, 30])
      b = np.array([1, 2, 3])
      print(a - b)
```

```
[ 9 18 27]
```

9. Counting values greater than 5

Without NumPy

```
[44]: a = [2, 7, 4, 9, 1, 8]
      count = 0

      for x in a:
          if x > 5:
              count += 1

      print(count)
```

```
3
```

With NumPy

[45]:
```
a = np.array([2, 7, 4, 9, 1, 8])
print(np.sum(a > 5))
```

```
3
```

10. Element-wise division

Without NumPy

[46]:
```
a = [10, 20, 30, 40]
b = [2, 4, 5, 8]
result = []

for i in range(len(a)):
    result.append(a[i] / b[i])

print(result)
```

```
[5.0, 5.0, 6.0, 5.0]
```

With NumPy:

[47]:
```
import numpy as np

a = np.array([10, 20, 30, 40])
b = np.array([2, 4, 5, 8])

print(a / b)
```

```
[5. 5. 6. 5.]
```